On the Invariant Theory of Forms in n Variables

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§ 3. The Symbolic Identities[1].

We must now carry over the second fundamental theorem of the symbolic method: namely, determine the totality of the irreducible and indecomposable identities in which all rows S^ρ, p_ρ are regarded as indecomposable. We make the following additional convention. In keeping with the meaning of symbol rows and variable rows, identities are to be regarded as composite when they arise by specializing the variable row p_ρ to $p_{\rho-1}y$ and applying one of the identities of the system to the resulting expressions. On the other hand, identities containing k symbol rows are to count as indecomposable even when the above process produces them from identities with fewer symbol rows. The rows S^ρ, p_ρ need not enter the identities explicitly; to regard them as indecomposable quantities, it is enough to show that the matrix products occurring depend only on these rows. We shall show, however, that in this case an explicit representation can always be obtained, partly by applying substitution (11). By the principle of matrix representation, we obtain the general identities from the special identities given by Pascal, which contain only the rows x and u and directly carry over those given by Study for the ternary domain:

$$\sum_{i=1}^{n}(-1)^{i+1}(a^{(i)} \mid x)(a^{(1)}\cdots a^{(i-1)}a^{(i+1)}\cdots a^{(n)}u) = (-1)^{n+1}(u \mid x)(a^{(1)}\cdots a^{(n)}), \tag{13}$$

$$\sum \pm(a^{(1)} \mid x^{(1)})(a^{(2)} \mid x^{(2)})\cdots(a^{(n)} \mid x^{(n)}) = (a^{(1)}\cdots a^{(n)})(x^{(1)}\cdots x^{(n)}), \tag{14}$$

$$\sum_{i=1}^{n}(-1)^{i+1}(u \mid a^{(i)})(a^{(1)}\cdots a^{(i-1)}a^{(i+1)}\cdots a^{(n)}x) = (-1)^{n+1}(u \mid x)(a^{(1)}\cdots a^{(n)}). \tag{15}$$

Two further identities result from (13) and (15) through the substitutions $(a^{(i)} \mid x) = (a^{(i)}q_{n-1})$ and $(u \mid a^{(i)}) = (p_{n-1}a^{(i)})$, respectively, and therefore are not, in our sense, to be regarded as essentially different. Identity (14) is the product theorem for determinants. We shall show how (13) and (14) (and, dually, (15) and (14)) can be replaced by a system of suitably formed product theorems for matrices. Applied once to ρ-row matrices and once to $(\rho-1)$-row matrices, the product theorem reads

$$(a^{(1)}a^{(2)}\cdots a^{(\rho)} \mid xp_{\rho-1}) = (a^{(1)}a^{(2)}\cdots a^{(\rho)} \mid xy^{(2)}\cdots y^{(\rho)}) = \sum \pm(a^{(1)} \mid x)(a^{(2)} \mid y^{(2)})\cdots(a^{(\rho)} \mid y^{(\rho)}),$$

$$\sum \pm(a^{(2)} \mid y^{(2)})\cdots(a^{(\rho)} \mid y^{(\rho)}) = (a^{(2)}\cdots a^{(\rho)} \mid y^{(2)}\cdots y^{(\rho)}) = (a^{(2)}\cdots a^{(\rho)}p_{\rho-1}) = (a^{(2)}\cdots a^{(\rho)}q_{n-\rho+1});$$

and from this follows the system of product theorems, where $\rho = 2, 3, \ldots, n$:

$$\begin{aligned}(a^{(1)}a^{(2)}\cdots a^{(\rho)} \mid xp_{\rho-1}) &= \sum_{i=1}^{\rho}(-1)^{i+1}(a^{(i)} \mid x)(a^{(1)}\cdots a^{(i-1)}a^{(i+1)}\cdots a^{(\rho)} \mid p_{\rho-1}) \\ &= \sum_{i=1}^{\rho}(-1)^{i+1}(a^{(i)} \mid x)(a^{(1)}\cdots a^{(i-1)}a^{(i+1)}\cdots a^{(\rho)}q_{n-\rho+1}).^{2}\end{aligned} \tag{16}$$

For $\rho = n$, (16) becomes (13). If we specialize further, setting $p_{n-1} = y^{(2)}p_{n-2}$, $p_{n-2} = y^{(3)}p_{n-3}$, and so forth, and apply the identities (16) to the expressions $(a^{(1)}\cdots a^{(i-1)}a^{(i+1)}\cdots a^{(n)} \mid y^{(2)}p_{n-2})$, $(a^{(1)}\cdots a^{(i-1)}a^{(i+1)}\cdots a^{(h-1)}a^{(h+1)}\cdots a^{(n)} \mid y^{(3)}p_{n-3})$, and so forth, then we pass from (13) to (14). Thus, by our definition, identity (14) is composite from the identities (16); equivalently, the system (16) is equivalent to identities (13) and (14). The system (16) satisfies one of the conditions required of the general identities: it contains the row $p_{\rho-1}$ as an indecomposable quantity and at the same time in explicit form. It is the only system satisfying this requirement, and this requirement alone. For no further determinants or matrix

[1]For the formulation and literature of the problem, cf. the Introduction.

[2]The identities (16) can also be regarded as the identities (13) carried over from the domain of ρ variables.

products can be formed from the rows a, x, p; nor can there be any relations among them independent of (16), since otherwise eliminating $(a^{(1)} \cdots a^{(\rho)} \mid xp_{\rho-1})$ would give a relation among ρ $(\rho \leq n)$ expressions

$$(a^{(i)} \mid x)(a^{(1)} \cdots a^{(i-1)} a^{(i+1)} \cdots a^{(\rho)} q_{n-\rho+1}),$$

whereas by (13) at least $n+1$ such expressions must enter into a relation.[3] Analogous considerations give the system equivalent to (15) and (14):

$$\begin{aligned}(uq_{\rho-1} \mid a^{(1)} \cdots a^{(\rho)}) &= \sum_{i=1}^{\rho} (-1)^{i+1} (u \mid a^{(i)})(q_{\rho-1} \mid a^{(1)} \cdots a^{(i-1)} a^{(i+1)} \cdots a^{(\rho)}) \\ &= \sum_{i=1}^{\rho} (-1)^{i+1} (u \mid a^{(i)})(p_{n-\rho+1} a^{(1)} \cdots a^{(i-1)} a^{(i+1)} \cdots a^{(\rho)}).\end{aligned} \tag{17}$$

To pass from (16) to identities among arbitrary rows $A^{\rho_1}, B^{\rho_2}, \ldots, x, p$, observe that these identities must become identical with (16) as soon as we resolve $A^{\rho_1} = a^{(1)} a^{(2)} \cdots a^{(\rho_1)}$, $B^{\rho_2} = b^{(1)} b^{(2)} \cdots b^{(\rho_2)}$, and so forth; moreover, they can be combined into final expressions only through the operations determined by (16), once the individual expressions are of the type occurring in (16). Thus, putting

$$\rho = \rho_1 + \rho_2 + \cdots + \rho_k \leq n, \tag{18}$$

we obtain

$$\begin{aligned}&(a^{(1)} \cdots a^{(\rho_1)} b^{(1)} \cdots b^{(\rho_2)} \cdots k^{(1)} \cdots k^{(\rho_k)} \mid xp_{\rho-1}) = (A^{\rho_1} B^{\rho_2} \cdots K^{\rho_k} \mid xp_{\rho-1}) \\ &= \sum_{i=1}^{\rho_1} (-1)^{i+1} (a^{(i)} \mid x)(a^{(1)} \cdots a^{(i-1)} a^{(i+1)} \cdots a^{(\rho_1)} B^{\rho_2} \cdots K^{\rho_k} q_{n-\rho+1}) \\ &\quad + (-1)^{\rho_1 \rho_2} \sum_{i=1}^{\rho_2} (-1)^{i+1} (b^{(i)} \mid x)(b^{(1)} \cdots b^{(i-1)} b^{(i+1)} \cdots b^{(\rho_2)} A^{\rho_1} \cdots K^{\rho_k} q_{n-\rho+1}) \\ &\quad + \cdots \\ &\quad + (-1)^{(\rho_1 + \cdots + \rho_{k-1})\rho_k} \sum_{i=1}^{\rho_k} (-1)^{i+1} (k^{(i)} \mid x)(k^{(1)} \cdots k^{(i-1)} k^{(i+1)} \cdots k^{(\rho_k)} A^{\rho_1} \cdots (K-1)^{\rho_{k-1}} q_{n-\rho+1}).\end{aligned}$$

Each individual sum (but not merely a part of a sum) does in fact contain the rows $A^{\rho_1}, B^{\rho_2}, \ldots$ as indecomposable quantities, because it satisfies the necessary and sufficient conditions[4]

$$\left(a^{(i+1)} \middle| \frac{\partial}{\partial a^{(i)}}\right) = 0; \ldots ; \left(k^{(j+1)} \middle| \frac{\partial}{\partial k^{(j)}}\right) = 0. \quad (i = 1, 2, \ldots, \rho_1 - 1;\ j = 1, 2, \ldots, \rho_k - 1) \tag{19}$$

We show that each sum also contains all the rows explicitly. Indeed, if we put $B^{\rho_2} \cdots K^{\rho_k} q_{n-\rho+1} = q_{n-\rho_1+1}$, then by (16) and (10) we obtain

$$\begin{aligned}\sum (-1)^{i+1} (a^{(i)} \mid x)(a^{(1)} \cdots a^{(i-1)} a^{(i+1)} \cdots a^{(\rho_1)} q_{n-\rho+1}) &= (A^{\rho_1} \mid xp_{\rho_1-1}) \\ &= (A_{n-\rho_1} x p_{\rho_1-1}) = (-1)^{(\rho_1+1)(n+1)} (q_{n-\rho_1+1} \mid A_{n-\rho_1} x) \\ &= (-1)^{(\rho_1+1)(n+1)} (B^{\rho_2} \cdots K^{\rho_k} q_{n-\rho+1} \mid A_{n-\rho_1} x).\end{aligned}$$

Computing the remaining sums analogously and then putting $B^{\rho_2} = A^{\rho_2}$, $C^{\rho_3} = A^{\rho_3}, \ldots$—the weight indices already characterize the rows as distinct—we obtain the desired identities:

$$\begin{aligned}(A^{\rho_1} A^{\rho_2} \cdots A^{\rho_k} \mid xp_{\rho-1}) &= \sum_{i=1}^{k} (-1)^{\delta_i} (A^{\rho_1} \cdots A^{\rho_{i-1}} A^{\rho_{i+1}} \cdots A^{\rho_k} q_{n-\rho+1} \mid A_{n-\rho_i} x), \\ \delta_i &= (\rho_1 + \rho_2 + \cdots + \rho_{i-1})\rho_i + (\rho_i + 1)(n+1).\end{aligned} \tag{20}$$

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[3] The print has $\rho \leq n$; R823 and the inherited English witness lose the equality stroke. In the same prose repetition of the factor in (16), both the print and R823 omit the ellipsis between $a^{(i+1)}$ and $a^{(\rho)}$; it is restored here from (16).

[4] Capelli, loc. cit., § 23, gives sufficient conditions: $\left(a^{(k)} \mid \frac{\partial}{\partial a^{(i)}}\right) = 0$, $(k > i,\ i = 1, 2, \ldots, \rho_1 - 1)$. The necessary and sufficient conditions follow most simply by applying the Poisson bracket process after Wellstein, *Von den Differentialgleichungen der projektiven Invarianten*, Math. Ann. 67 (1909), § 3.

From (17) we obtain the dual identities:

$$(uq_{\sigma-1} \mid A_{\sigma_1}A_{\sigma_2}\cdots A_{\sigma_k}) = \sum_{i=1}^{k}(-1)^{\varepsilon_i}(uA^{n-\sigma_i} \mid p_{n-\sigma+1}A_{\sigma_1}\cdots A_{\sigma_{i-1}}A_{\sigma_{i+1}}\cdots A_{\sigma_k}),$$
$$\varepsilon_i = (\sigma_1+\sigma_2+\cdots+\sigma_{i-1})\sigma_i + (\sigma_i+1)(n+\sigma). \tag{21}$$

Theorem II. Equations (20) and (21) exhaust the totality of the indecomposable identities among the rows A^{ρ_i}, x, p, respectively A_{σ_i}, u, q, and at the same time the totality of those indecomposable identities that admit an explicit representation without carrying out substitution (11).

The first part of the theorem follows from the considerations leading to (20) and (21); the second follows from the impossibility of an identity between two symbol rows:

$$(q_\rho A^\sigma \mid B_{\rho+\sigma-\tau}p_\tau) = \varepsilon_1(q_\rho B^{n-\rho-\sigma+\tau} \mid A_{n-\sigma}p_\tau) + \varepsilon_2(q_\rho q_{n-\tau} \mid B_{\rho+\sigma-\tau}A_{n-\sigma}),$$

where $\rho \le \tau \le \sigma$ and none of the rows has weight $1\cdot(n-1)$. (Other assumptions about ρ, τ, σ reduce to this case by interchanging the row designations.) This impossibility follows, for example, by expanding (8) under the specialization $q_\rho = lM^{\rho-1}$, $q_{n-\tau} = lN^{n-\tau-1}$, if one sets $l_1 = 1$, $M^{2,3,\ldots,\rho} = 1$, $A^{\rho+1,\ldots,\rho+\sigma} = 1$, and all other elements of the rows l, M, A equal to zero, while N and B remain arbitrary. Since identities among arbitrary rows, after a row p_τ is resolved into $y^{(1)}y^{(2)}\cdots y^{(\tau)}$, pass into identities composite from (20), the second part of the assertion is proved once (20) can be generated from identities containing only two symbol rows. Indeed, from

$$(A^{\rho_1}A^{\rho_2} \mid xp_{\rho-1}) = \varepsilon(A^{\rho_1}q_{n-\rho+1} \mid A_{n-\rho_2}x) + (A^{\rho_1} \mid xp_{\rho_1-1}), \quad \text{where } p_{\rho_1-1} \sim A^{\rho_2}q_{n-\rho+1}, \tag{20a}$$

by specializing $A^{\rho_1} = B^{\sigma_1}B^{\sigma_2}$, that is, by using

$$(B^{\sigma_1}B^{\sigma_2} \mid xp_{\rho_1-1}) = \varepsilon(B^{\sigma_1}q_{n-\rho_1+1} \mid B_{n-\sigma_2}x) + (B^{\sigma_1} \mid xp_{\sigma_1-1}), \quad \text{where } p_{\sigma_1-1} \sim B^{\sigma_2}q_{n-\rho_1+1},$$

one obtains an identity (20) in three symbol rows; and further specializations $B^{\sigma_1} = C^{\tau_1}C^{\tau_2}$, and so forth, give the general identities (20). Thus the nonexistence of an explicit representation for an identity between two symbol rows and two arbitrary variable rows implies nonexistence for arbitrarily many symbol rows, provided that no row x or u occurs among the variable rows.